



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.1

Understand Ratios

Lesson 3-1 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can write and describe a ratio that compares two quantities.

Language Objective: I can explain a comparison using the words ratio, part-to-part, part-to-whole, and colon notation.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Ratio

Comparison

Part-to-part

Part-to-whole

Colon notation



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1 A comparison of two quantities, like 3 to 5, is a .

2 Showing how two amounts relate to each other is a .

3 A ratio that compares one part of a group to another part is a ratio.

4 A ratio that compares one part of a group to the total is a ratio.

5 Writing a ratio with a colon, like 3:5, is using .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

When you sorted the board statements, what made something a ratio instead of NOT a ratio (like '12 cookies on the tray')?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Ratio** — A way to compare two amounts, like 3 to 2.

Razón — Una manera de comparar dos cantidades, como 3 a 2.

- ☐ **Comparison** — Looking at two or more amounts to see how they are related.

Comparación — Mirar dos o más cantidades para ver cómo se relacionan.

- ☐ **Part-to-whole** — A ratio comparing one part to the whole group.

Parte a todo — Una razón que compara una parte con todo el grupo.

- ☐ **Colon notation** — Writing a ratio with two dots between the numbers, like 3:2.

Notación con dos puntos — Escribir una razón con dos puntos entre los números, como 3:2.

- ☐ **Part-to-part** — A ratio comparing one part of a group to another part.

Parte a parte — Una razón que compara una parte de un grupo con otra parte.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

This is a ratio because it compares ____ to ____.

Esto es una razón porque compara ____ con ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The recipe uses 3 cups of apple juice for every 2 cups of sparkling water.

Evidence: A strong answer names that apple juice and sparkling water are being compared and states the relationship as 3 to 2 (or 'for every 3 apple juice, 2 sparkling water'), not just that there is more apple juice.

Reasoning: This evidence connects the definition of ratio to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **This is a ratio because it compares ____ to ____.**

Esto es una razón porque compara ____ con ____.

- **This is NOT a ratio because it only ____.**

Esto NO es una razón porque solo ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Ratio
2. **Notes 2:** Comparison
3. **Notes 3:** Part-to-part
4. **Notes 4:** Part-to-whole
5. **Notes 5:** Colon notation
6. **Watch for this mistake:** *A common mistake in Understand Ratios is writing the ratio in the wrong order: when asked for the ratio of apple juice to sparkling water in a recipe that uses 3 cups of apple juice for every 2 cups of sparkling water, some students write 2:3 (sparkling water to apple juice) because they list the ingredients in whatever order comes to mind rather than the order the question asks for. The two ratios describe different drinks — 3:2 means 3 cups of juice for every 2 cups of water, while 2:3 would mean only 2 cups of juice for every 3 cups of water. Always match the order of your ratio to the order named in the question: "apple juice to sparkling water" means the apple juice number comes first, giving 3:2.*
7. **Try It 1:** A. 5:12 — *The whole class is $5 + 7 = 12$ students. Part-to-whole compares one part to the total, so chefs to whole class is 5:12.*
8. **Try It 2:** A. 9:4 — *"For every 9 honey to 4 lemon" is a part-to-part comparison. In colon notation, honey to lemon is 9:4.*